

E2 — Embedded Track

Member: _____

Your role

You own the measurement protocol and the bandwidth benchmark. You verify the memory and bandwidth claims on hardware.

Weekly tasks

Week 1 — Bandwidth benchmark study

- Task: Build the bandwidth benchmark firmware on host (compile only). Read firmware/benchmarks/bandwidth. Write the measurement protocol: tok/s, ms/token, memory free, bandwidth numbers.
- Deliverable: The measurement protocol doc.
- Verify: E1 reviews the protocol and finds no missing step.
- Learn: Memory hierarchy: SRAM vs PSRAM vs flash. Explain read latency.

Week 2 — Flash tutorial

- Task: Write the flashing tutorial: cables, ports, commands, expected serial output.
- Deliverable: FLASH_TUTORIAL.md in the club repo.
- Verify: H2 can flash a board from the tutorial alone.
- Learn: The deploy pipeline: fetch_model.sh, deploy.sh. Explain the SHA-256 pin check.

Week 3 — Measurement checklist

- Task: Write the measurement checklist for week 5: tools, steps, recording format.
- Deliverable: The checklist doc.
- Verify: E1 runs the checklist on paper without a question.
- Learn: Runtime internals: the int8 head, the PLE path. Explain why the head dominates the time.

Week 4 — Week-5 rehearsal

- Task: Rehearse the week-5 measurement steps on paper with E1. Finalize the flash plan.
- Deliverable: The signed-off flash plan.
- Verify: Both of you can state the first three steps from memory.
- Learn: Measurement protocol. Explain why you record 3 measurements per number.

Week 5 — Barista + bandwidth

- Task: Flash the stock barista model. Run the bandwidth benchmark firmware. Measure flash row read, PSRAM, SRAM speeds.
- Deliverable: The bandwidth verification table.
- Verify: Your numbers sit within +/-15% of published: 20.3 us, 60.7 MB/s, 240 MB/s.
- Learn: On-chip verification. Compare your table with the published table.

Week 6 — Stability + schedule

- Task: Run the stability tests: power cycles, boot loop, long-run generation. Own the board schedule with E1.
- Deliverable: The stability report and the board schedule.
- Verify: The board survives 10 power cycles.
- Learn: Firmware features: serial prompting, dual-core hooks. Read the layer_matvec hook code.

Week 7 — Demo rehearsal

- Task: Run the live demo rehearsal on hardware.
- Deliverable: The rehearsal report with timings.
- Verify: The demo runs start to finish without help.
- Learn: Demo operations: power, boot, serial. Write the demo checklist.

Week 8 — Final on-chip numbers

- Task: Record the final on-chip numbers for the record.
- Deliverable: ON_CHIP_FINAL.md in the club repo.
- Verify: Every number matches the tracking sheet.
- Learn: Write the "what I learned" page.

Learning path

1. ESP32-S3 architecture: cores, SRAM, PSRAM, flash.
2. Deploy pipeline and boot flow.
3. Runtime internals: int8 head, KV cache, memory budget.
4. Measurement protocol: tok/s, ms/token, bandwidth.
5. On-chip verification vs published numbers.
6. Firmware features: serial prompting, profiling.
7. Demo hardening: power, boot, serial reliability.
8. Live demo and final numbers.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- firmware/benchmarks/bandwidth: <https://github.com/slvDev/esp32-ai/tree/main/firmware/benchmarks/bandwidth>
- RESULTS.md: <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- ESP32-S3 docs: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32s3/>